

# Optimization

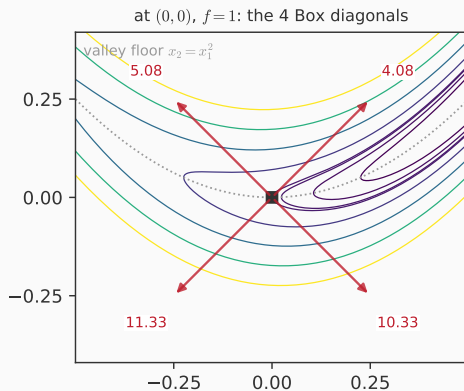
## 3.2.5 Hooke–Jeeves method (pattern search)

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José Firmino Aguilar Madeira

# What if, instead of the diagonals, we try the axes?

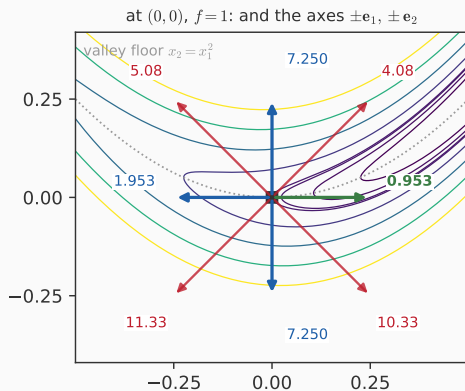
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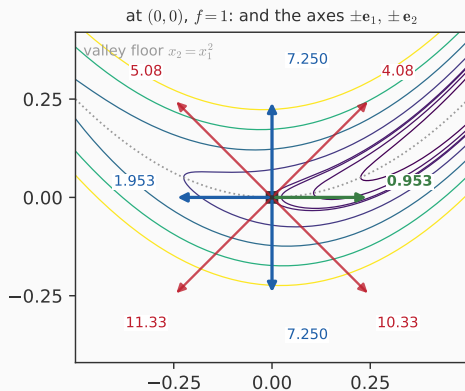
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Why not test  $\pm e_1, \dots, \pm e_n$ ?

| $n$ | Box: $2^n$     | axes: at most $2n$ |
|-----|----------------|--------------------|
| 2   | 4              | 4                  |
| 10  | 1 024          | 20                 |
| 20  | $\approx 10^6$ | 40                 |



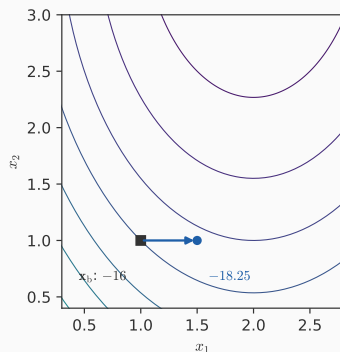
# One exploration by hand

$f(x_1, x_2) = 3x_1^2 + x_2^2 - 12x_1 - 8x_2$  (minimum at  $(2, 4)$ ,  
 $f = -28$ ).

Base point  $x_b = (1, 1)$ ,  $f = -16$ ; perturbations

$P = (0.5, 0.5)$ .

1. **Coordinate**  $x_1$ :  $(1.5, 1)$  gives  $-18.25 < -16$  ✓ We keep it.



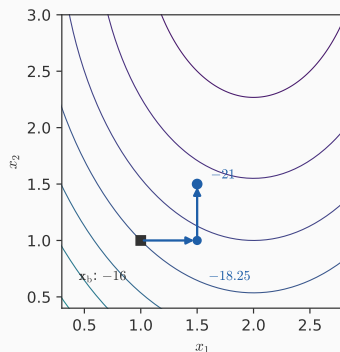
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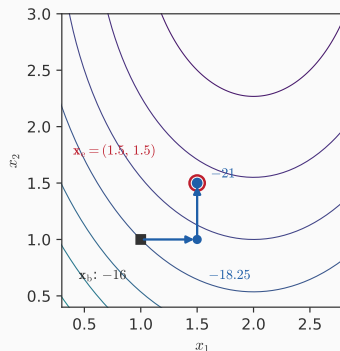
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2. **Coordinate**  $x_2$ , from the **already improved** point:  $(1.5, 1.5)$  gives  $-21$  ✓
3. Result:  $x_b = (1, 1) \rightarrow x_e = (1.5, 1.5)$ , with 2 evaluations.

The coordinates are explored **sequentially**: each trial starts from the **best point found so far**.



# Exploratory move

## Exploration around $x_b$ with perturbations

$P = (P_1, \dots, P_n)$

```
1:  $x \leftarrow x_b$ ;  $f_{\text{best}} \leftarrow f(x_b)$ 
2: for  $j = 1, \dots, n$  do
3:   try  $x_j + P_j$ : if it improves  $f_{\text{best}}$ , keep it and go to
   the next  $j$ 
4:   else try  $x_j - P_j$ : if it improves, keep it
5:   else leave  $x_j$  as it was
6: end for
7: return  $x_e \leftarrow x$  ▷ failed if  $x_e = x_b$ 
```

$n \leq n_f \leq 2n$  per exploration

- $n_f = n$  if all the  $+P_j$  trials work;  $2n$  if all fail.
- **Box:** all combinations of  $\pm (2^n)$ .  
**Hooke–Jeeves:** one coordinate at a time (up to  $2n$ ).
- The direction obtained,  $x_e - x_b$ , has components in  $\{-P_j, 0, +P_j\}$ : a rough indication of the direction of progress (not the best one), without derivatives.

## We have a direction that worked: why not extrapolate?

The exploration took us from  $x_b$  to  $x_e$  and improved. We have information that Box did not exploit:

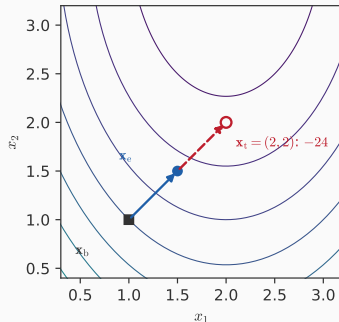
$$d = x_e - x_b.$$

Why explore again only around  $x_e$ ? If that direction worked:

we go further in the same direction

$$x_t = x_b + a(x_e - x_b), \quad a = 2: \quad x_t = 2x_e - x_b.$$

In the example:  $x_t = 2(1.5, 1.5) - (1, 1) = (2, 2)$ ,  $f = -24$ .



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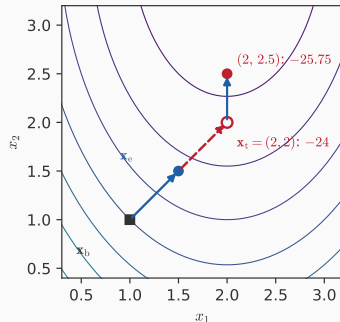
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And we **explore again** around  $x_t$ :  $(2, 2) \rightarrow (2, 2.5)$ ,  
 $f = -25.75 < -21$  — the jump is accepted.



# Exploratory + pattern: the Hooke–Jeeves method

Hooke and Jeeves (1961): the two moves we have just built, alternated.

## Exploratory move

**Local** search, axis by axis, from the base point  $x_b$ : perturb each coordinate by  $\pm P_j$  and keep whatever improves. Result: the explored point  $x_e$ . Cost:  $n$  to  $2n$ .

## Pattern move

**Extrapolation** in the direction of progress: trial point  $x_t = x_b + a(x_e - x_b)$ , followed by a new exploration. Repeated while there is progress: the jumps *accumulate* the direction that works.

the pattern **accelerates**

the exploration **corrects**

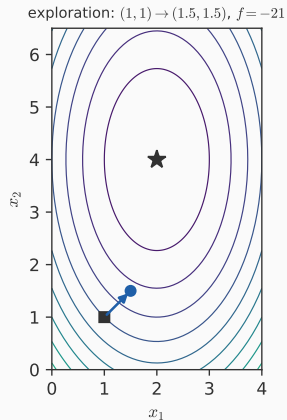
The exploratory steps are aligned with the axes, but extrapolation allows following oblique valleys.

The perturbations  $P_j$  are the *poll step*: they play the role of the step  $\alpha_k$  of the other methods (and return in 5.1 and 6.5).

# Complete example: the path curves

$x_0 = (1, 1)$ ,  $a = 2$ ,  $P_0 = (0.5, 0.5)$ ,  $T = (0.1, 0.1)$  (large tolerance, just so the example ends quickly).

1. Exploration at  $(1, 1)$ ,  $f = -16$ :  $(1.5, 1) \rightarrow -18.25 \checkmark$ ;  
 $(1.5, 1.5) \rightarrow -21 \checkmark$ .  $x_e = (1.5, 1.5)$ .

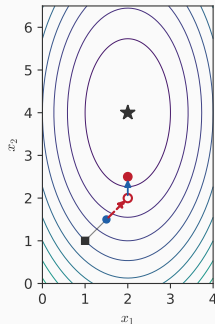


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2. **Pattern:** trial  $2(1.5, 1.5) - (1, 1) = (2, 2)$ ,  $f = -24$ ; exploration  
 $\rightarrow (2, 2.5)$ ,  $f = -25.75 < -21$ : accepted.

pattern  $(2, 2)$ , explor.  $\rightarrow (2, 2.5)$ ,  $f = -25.75$ : accepted

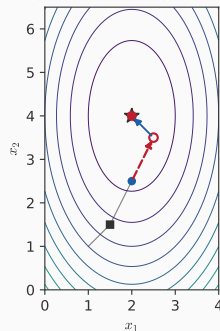


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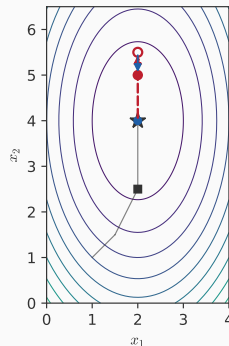


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 $\rightarrow (2, 4)$ ,  $f = -28$ : **accepted**.
4. **Pattern**: trial  $2(2, 4) - (2, 2.5) = (2, 5.5)$ ,  $f = -25.75$ ; exploration  
 $\rightarrow (2, 5)$ ,  $f = -27 \geq -28$ : **rejected**. Back to  $(2, 4)$ .

pattern  $(2, 5.5)$ , explor.  $\rightarrow (2, 5)$ ,  $f = -27$ : rejected

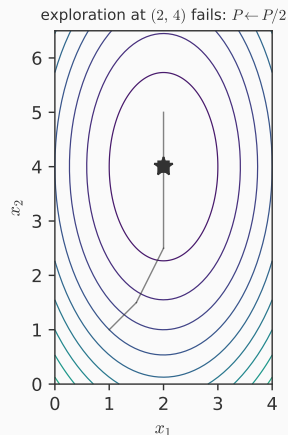


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 $\rightarrow (2, 5)$ ,  $f = -27 \geq -28$ : **rejected**. Back to  $(2, 4)$ .
5. Exploration at  $(2, 4)$  fails with  $P = 0.5$ ; with  $0.25$ ; with  $0.125$ ;  
 $0.0625 < T$ : stop. Solution  $(2, 4)$ ,  $f = -28$ .

The pattern always points in a straight line; the exploration corrects; the path **curves**.



# The algorithm

explore around  $x_b$   
fails  $\Rightarrow P \leftarrow P/2$   
improves  $\Rightarrow$  **extrapolate** ( $x_t$ )  
 $\downarrow$   
**explore** again around  $x_t$   
improved? yes  $\Rightarrow$  continue the  
pattern  
no  $\Rightarrow$  step back and restart

## Rule of thumb

P **only decreases**: do not reset  $P_0$  at each success.

On Rosenbrock, resetting  $P_0$  after each successful exploration:  $n_f = 21\,246$  until  $f < 10^{-4}$ , against  $353$  ( $\approx 60\times$ ).

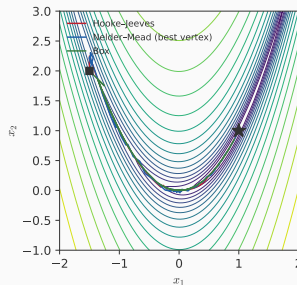
**Data:**  $x_0$ , factor  $a$ , initial perturbations  $P_0$ , tolerances  $T$

```
1:  $P \leftarrow P_0$ ;  $x_b \leftarrow x_0$ 
2: loop ▷ start / restart
3:    $x_e \leftarrow$  exploration around  $x_b$ 
4:   if failed then
5:      $P \leftarrow P/2$ ; if all  $P_j < T_j$ : return  $x_b$ 
6:   else
7:     loop ▷ pattern moves
8:        $x_t \leftarrow x_b + a(x_e - x_b)$  ▷ trial point
9:        $x'_e \leftarrow$  exploration around  $x_t$ 
10:      if  $f(x'_e) \geq f(x_e)$  then  $x_b \leftarrow x_e$ ; exit (restart)
11:      else  $x_b \leftarrow x_e$ ;  $x_e \leftarrow x'_e$ 
12:    end if
13:  end loop
14: end if
15: end loop
```

Stopping: the exploration fails at all scales down to  $T$ , in *all* coordinates. Cost: each pattern move spends  $1 + (n \text{ to } 2n)$  evaluations (the trial point and the new exploration).

# Where it fails, and the four methods on Rosenbrock

- **The stopping test can be fooled:** if a narrow valley passes between the points  $x \pm P_j e_j$ , the exploration fails along all axes although a descent direction exists. *Possible heuristic:* add a few random directions — it helps, but does not solve the problem of directional coverage.
- **Guarantee:** the coordinate exploration of HJ is the origin of the pattern search framework; formalized versions such as GPS (Torczon, 1997) make it possible to establish stationarity results under suitable assumptions (e.g.  $f \in C^1$  and bounded level set  $\Rightarrow \liminf \|\nabla f(x_k)\| = 0$ ). With a finite step, stopping does not certify stationarity: it only says that none of the  $2n$  directions improved with that step. HJ already uses a positive basis ( $\pm e_j$ ); modern pattern search generalizes it.
- In curved valleys it zigzags: successful pattern, failure, new exploration with a reduced step.



| Rosenbrock from $(-1.5, 2)$            | $n_f$ to $f < 10^{-4}$ |
|----------------------------------------|------------------------|
| Nelder-Mead (edges 0.5)                | 166                    |
| Hooke-Jeeves ( $P_0 = 0.5$ , $a = 2$ ) | 353                    |
| localized random <sup>†</sup>          | 975 [867, 1052]        |
| Box ( $\Delta_0 = 1$ )                 | 13 445                 |

<sup>†</sup> median [Q1, Q3], 30 runs. On this problem, with these parameters and this criterion.

# On the computer: from Hooke–Jeeves to modern pattern search

The core, in MATLAB:

```
for j = 1:n                                % exploration (xe = copy of xb)
    xp = xe; xp(j) = xp(j) + P(j); fp = f(xp); % xp: trial
    if fp >= fe
        xp(j) = xe(j) - P(j); fp = f(xp);
    end
    if fp < fe, xe = xp; fe = fp; end
end
xt = xb + a*(xe - xb);                    % pattern: trial point
```

The full functions (Exploratory, HookeJeeves, with info.nfev) are in the course companion code.

In practice, the variables should be scaled and  $P_{0,j}$  and  $T_j$  chosen relative to the scale of each variable (one may be in  $10^{-3}$  m, another in  $10^5$  Pa).

## patternsearch (MATLAB)

A modern pattern search (positive bases with  $2n$  or  $n + 1$  directions), which generalizes classical Hooke–Jeeves, with convergence theory (GPS) under suitable assumptions.

Hooke–Jeeves (the origin) → modern  
pattern search

→ directional direct search

(poll, local component of GLODS, 5.1)

→ DMS, multiobjective (6.5)

## Summary table — 3.2

| Method               | Information   | Cost                                          | Guarantee                                                                                                                         | Main limitation             |
|----------------------|---------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| Random (3.2.1)       | values of $f$ | 1 per sample                                  | global, in probability, as $N \rightarrow \infty$ (full support, $f$ continuous)                                                  | $\ x - x^*\  \sim N^{-1/n}$ |
| Grid (3.2.2)         | values of $f$ | $m^n$ in total                                | $f_{\text{grid}} \rightarrow f^*$ ( $f$ continuous);<br>$f_{\text{grid}} - f^* \leq L \frac{\sqrt{n}}{2} h$ ( $f$ $L$ -Lipschitz) | exponential in $n$          |
| Nelder–Mead (3.2.3)  | values of $f$ | 1–2 per iteration<br>( $n + 2$ if it shrinks) | no general one (McKinnon, 1998)                                                                                                   | may stagnate                |
| Box (3.2.4)          | values of $f$ | $2^n$ per iteration                           | no general guarantee presented in this course; a small box does not certify stationarity at finite $\Delta$                       | $2^n$ ; only diagonals      |
| Hooke–Jeeves (3.2.5) | values of $f$ | $n-2n$ per exploration                        | formalized versions (GPS): stationarity in the limit, under suitable assumptions                                                  | axes; zigzag                |

random + grid = **explore**

Nelder–Mead + Box + Hooke–Jeeves = **refine locally**

NM has no general guarantee; HJ is the origin of pattern search, whose formalized versions (GPS) have stationarity in the limit under suitable assumptions (3.2.5); for Box we present no general guarantee. Only evaluations of  $f$ : the direction is inferred, and that costs evaluations; derivative-free  $\neq$  robust to noise.

## KEY TAKEAWAYS — 3.2.5

1. Hooke–Jeeves alternates axis-by-axis **exploration** ( $\pm P_j$  in each coordinate,  $n$  to  $2n$  evaluations) and **pattern move** (extrapolating in the direction of progress).
2. The pattern **accelerates** and the exploration **corrects**: the path curves and follows oblique valleys, but zigzags in curved valleys.
3. The perturbations  $P$  only decrease, and only after an unsuccessful exploration.
4. The coordinate exploration of HJ is the origin of pattern search; formalized versions (GPS) have **stationarity in the limit, under suitable assumptions**; with a finite step, stopping does not certify stationarity.
5. Hooke–Jeeves  $\rightarrow$  modern pattern search  $\rightarrow$  GLODS poll (5.1)  $\rightarrow$  DMS (6.5).

Next (3.3): so far we have inferred directions only through evaluations of  $f$ , and that costs evaluations. **What if  $f$  is smooth and we have  $\nabla f$ ?** — derivative-based methods.

**References** Full list: *Course bibliography*, at the end of the course PDF.

- Hooke, R., Jeeves, T. A., “*Direct search*” solution of numerical and statistical problems, Journal of the ACM 8(2):212–229, 1961.
- Torczon, V., *On the convergence of pattern search algorithms*, SIAM Journal on Optimization 7(1):1–25, 1997.
- Conn, A. R., Scheinberg, K., Vicente, L. N., *Introduction to Derivative-Free Optimization*, SIAM (MPS-SIAM Series on Optimization 8), 2009 — Ch. 7.

## Resources

- Companion code: `Exploratory`, `HookeJeeves` (`info.nfev`). MATLAB: `patternsearch` (Global Optimization Toolbox).